

GENERATOR

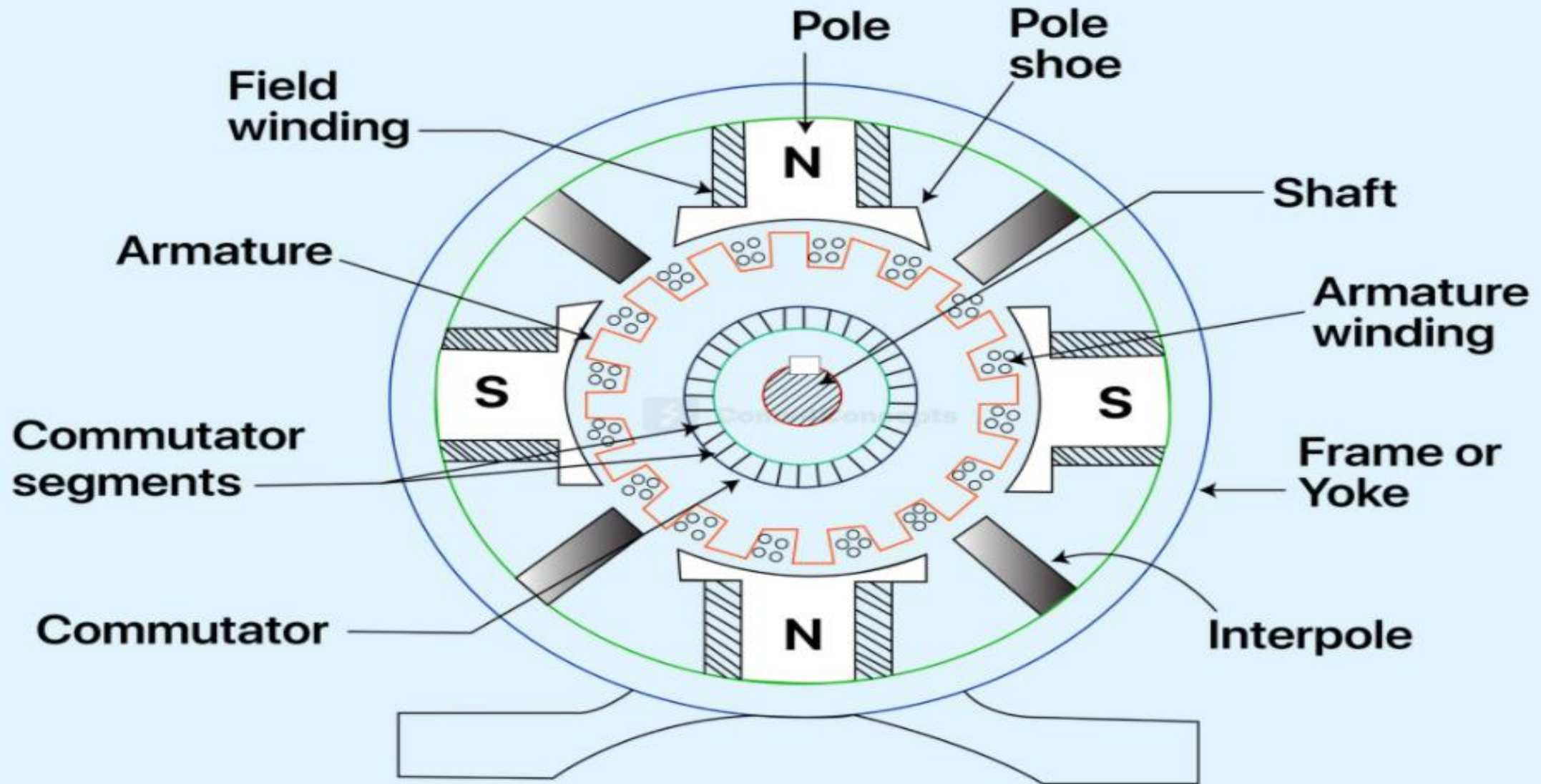
Converts Mechanical energy into electrical energy

Two types

Based on the output generated.

1. AC Generator
2. DC Generator

Construction



DC Generator

Construction

Yoke

- Outer frame made of cast iron and steel
- Provides Mechanical support to the generator
- Carries the magnetic flux that is generated

Pole core and pole shoes

- Field windings are connected in the pole core hence the pole core carries the field winding.
- Made up of cast iron or mild steel
- Pole shoes supports the windings wound on the pole core and also helps in distributing the generated flux uniformly into the machine

Interpoles

- Narrow poles fixed exactly midway between the main poles
- They are also called commutating poles.
- It helps in improving the commutation process

Field Winding

- Made up of copper
- To produce magnetic field 2 types of magnets can be used Permanent magnet or electro magnet
- Electro magnets are usually used.

Armature

- Armature core is cylindrical type with slots
- It is laminated to reduce the eddy current
- Made up of steel
- Armature conductors are wound in this slots.

Armature Windings

- Copper coils are used
- The conductors are separately insulated
- Armature winding are of two types
 1. Lap winding
 2. Wave winding

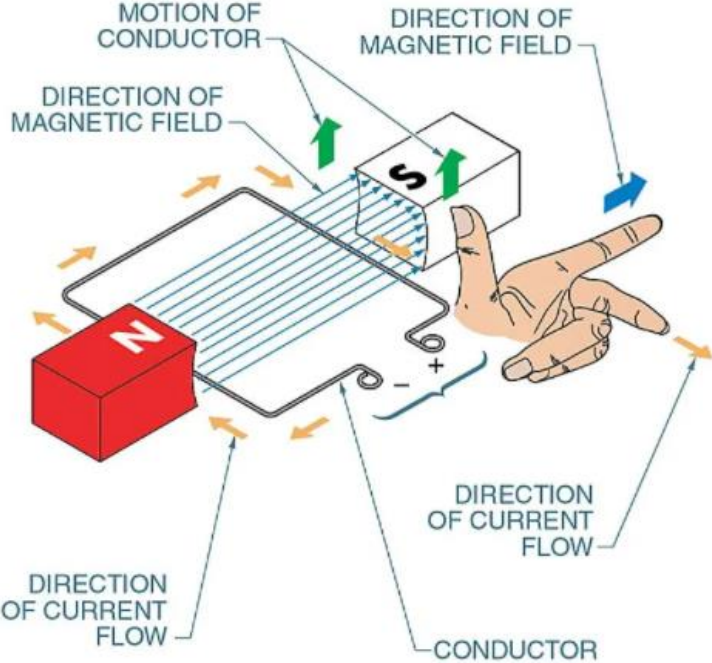
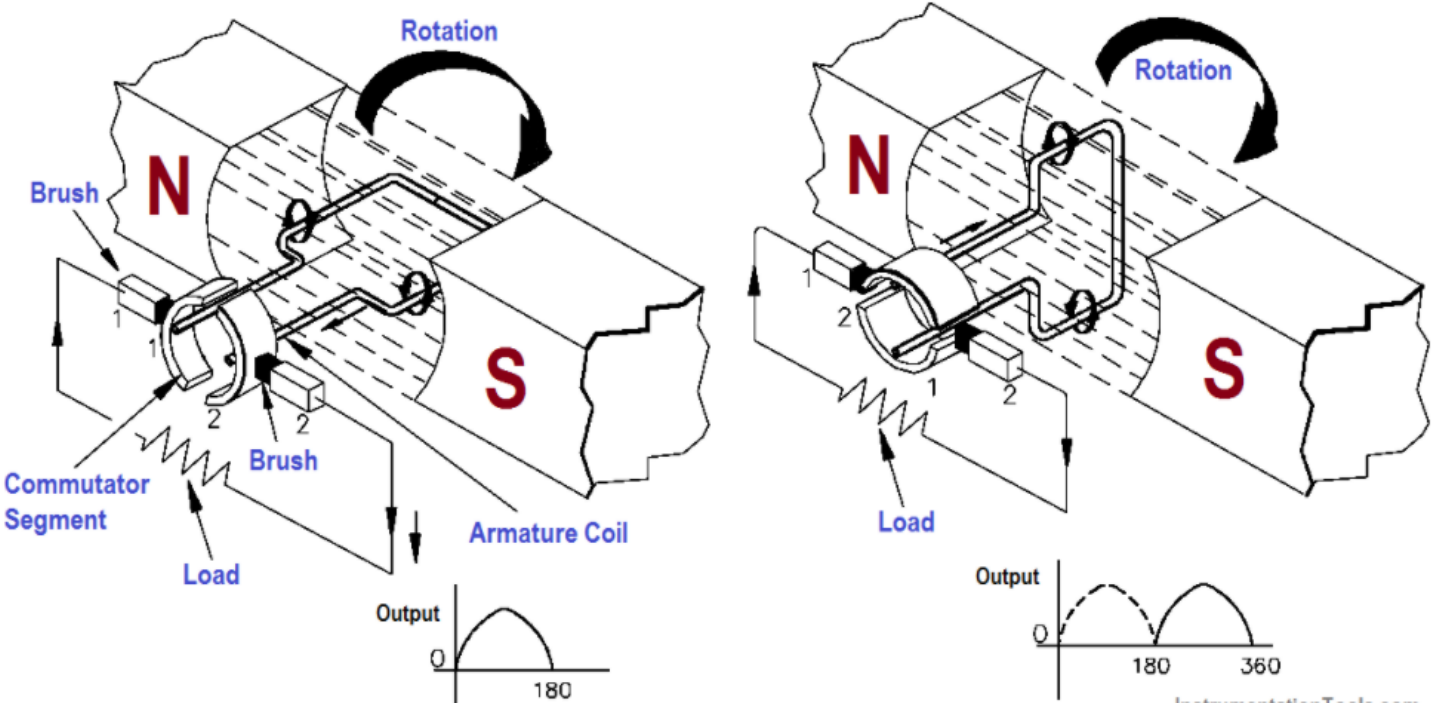
Commutator and brushes

- Generator o/p is DC
- Commutator converts AC to DC
- Split rings are used as commutator
- Brushes are used to collect the generated current it is made up of carbon or graphite

Shaft

- The mechanical rod on which the armature rotates

Working



Types of DC Generators

Permanent Magnet

Separately Excited

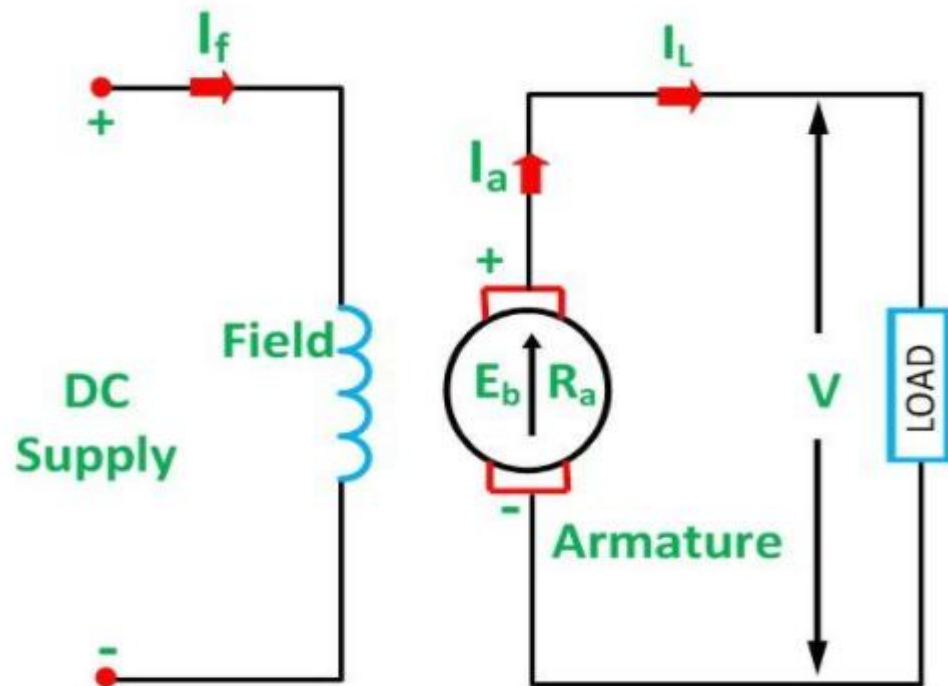
Self Excited

1) Series

2) Shunt

3) Compound

Separately Excited DC Generator



$$I_a = I_L = I$$

$$V = E_g - I_a R_a$$

$$V = E_g - I R_a$$

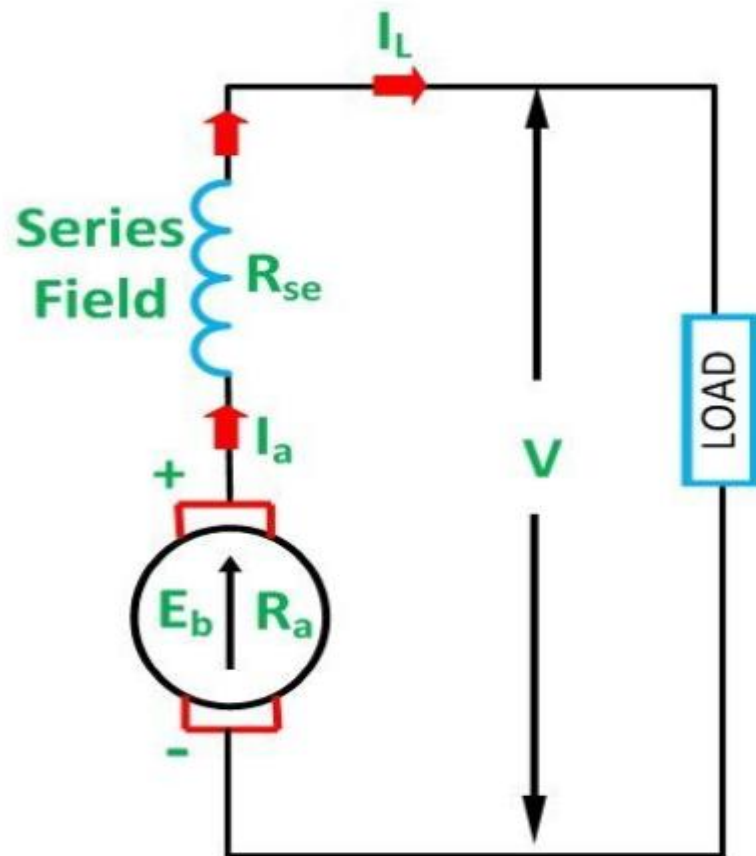
Self Excited DC Generator

Series

Shunt

Compound

Series Wound



Here,

$$I_a = I_{se} = I_L = I$$

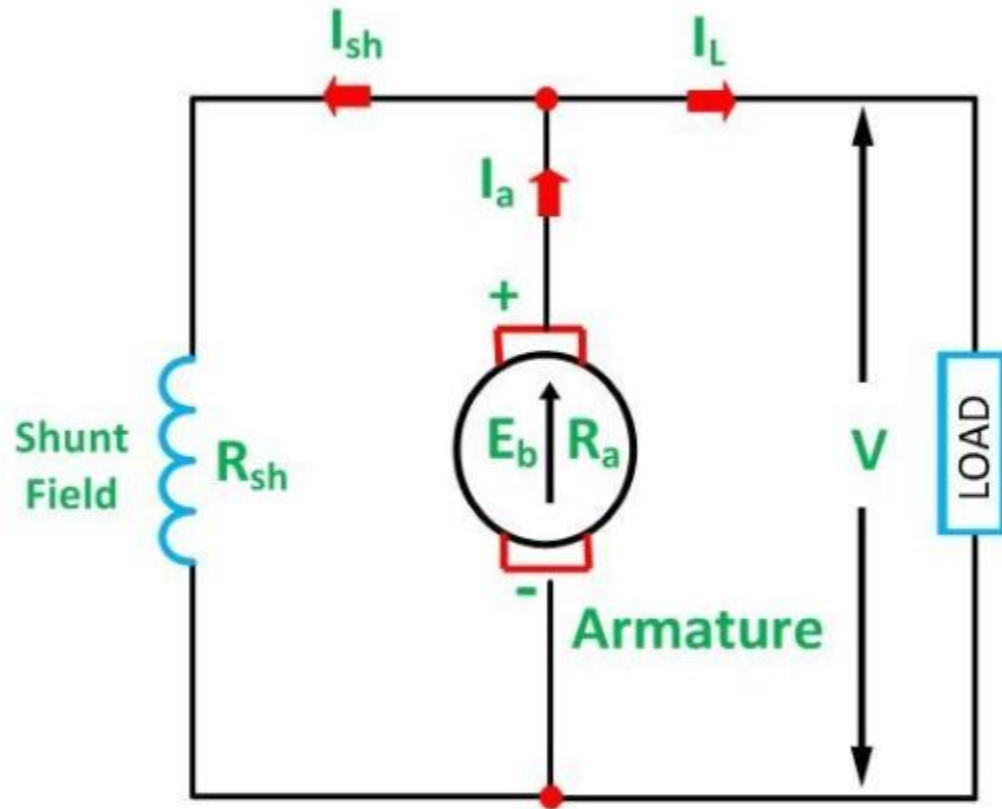
Considering drop across series winding,

$$V = E_g - I_a R_a - I_{se} R_{se}$$

So,

$$\mathbf{V = E_g - I (R_a + R_{se})}$$

Shunt Wound



Here,

$$I_a = I_{sh} + I_L$$

And,

$$I_{sh} = V/R_{sh}$$

$$V = E_g - I_a R_a$$

So,

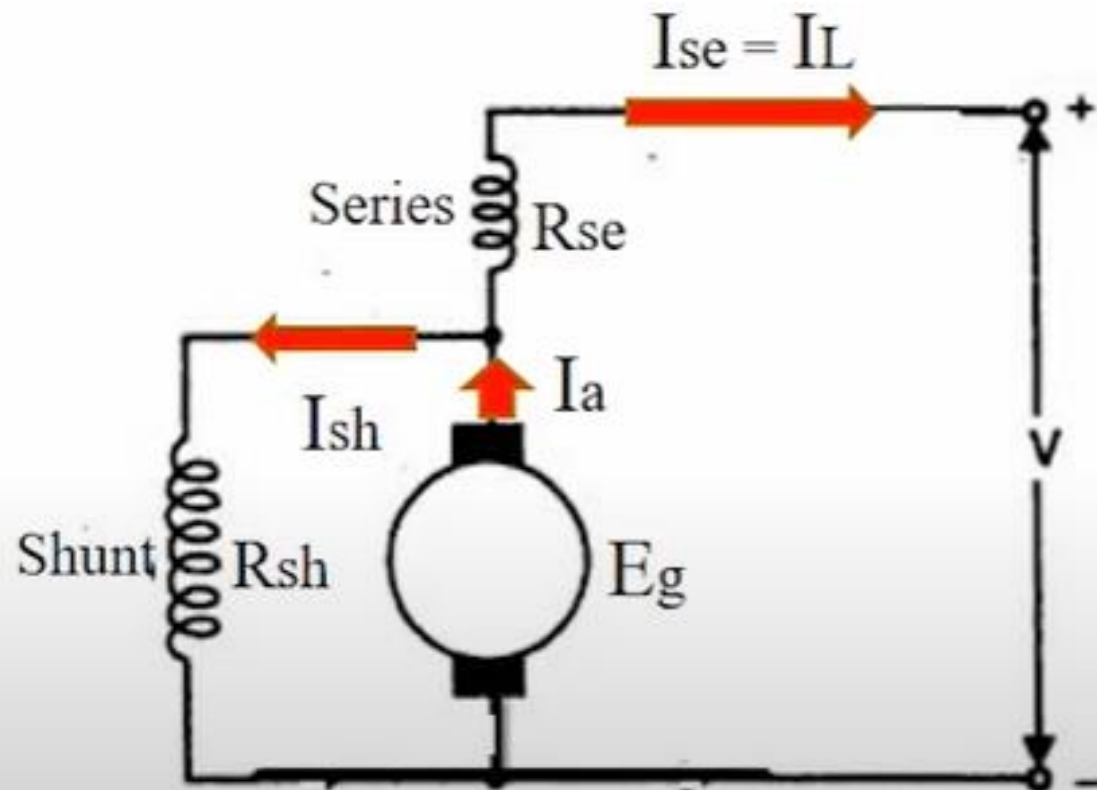
$$V = E_g - (I_{sh} + I_L) R_a$$

Compound wound DC Generator

2 type

- Short shunt
- Long shunt

Short Shunt



$$I_{se} = I_L$$

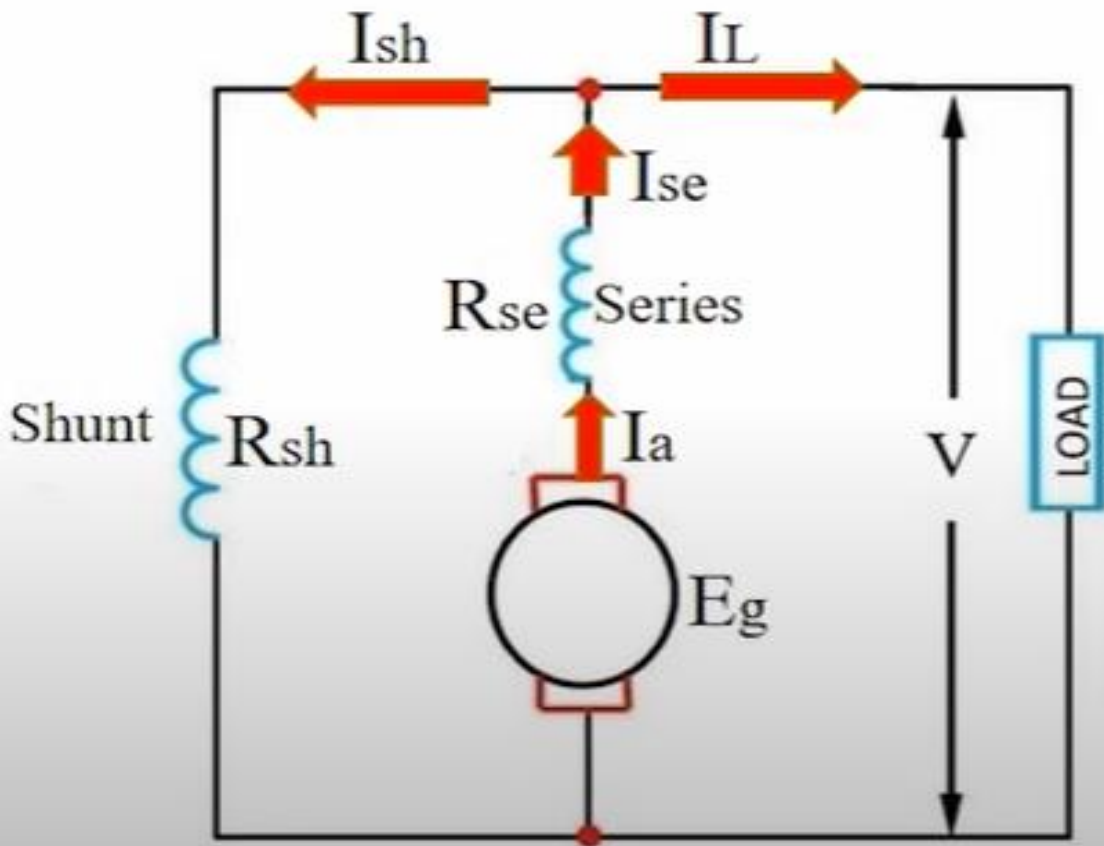
$$\text{And, } I_a = I_{sh} + I_{se}$$

$$I_{sh} = (V + I_{se}R_{se})/R_{sh}$$

So,

$$\mathbf{V = E_g - I_a R_a - I_{se} R_{se}}$$

Long Shunt



$$I_a = I_{se}$$

$$I_{sh} = V/R_{sh}$$

Also,

$$I_a = I_{se} = I_{sh} + I_L$$

$$V = E_g - I_a R_a - I_{se} R_{se}$$

So,

$$\mathbf{V = E_g - I_a (R_a + R_{se})}$$

DC MOTOR

- Converts mechanical energy into electrical energy
- Requires
 1. Field
 2. Conductor
 3. A current should be passed through the conductor
- **When a current carrying conductor is placed in a magnetic field , it experiences a mechanical force whose direction is given by Flemings left hand rule and magnitude, $F = BIL$ newton**
- There is no basic difference between generator and motor in their constructional details
- The function of the commutator in the DC motor is the same as in case of a generator, by reversing current in each conductor as it passes from one pole to another, it helps to develop a continuous and unidirectional torque.

SIGNIFICANCE OF BACK EMF

When the armature rotates, the conductor also rotates and hence cut the flux.

EMF is induced in them whose direction was found by Flemings right hand rule , it is in opposition to the applied voltage

Because of its opposite direction it is termed as back emf E_b

$$I_a = \frac{V - E_b}{R_a}, \quad R_a, \text{ is the armature resistance}$$

$$I_a R_a = V - E_b$$

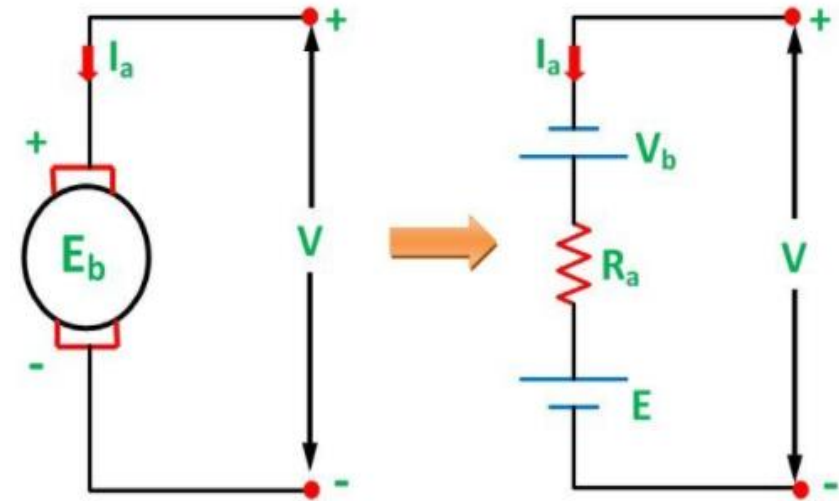
$$V = E_b + I_a R_a$$

Flux cut by one conductor = $P\phi$

$$\text{Time taken for one revolution} = \frac{60}{N}$$

$$E_b = \frac{P\phi}{\frac{60}{N}} = \frac{P\phi N}{60}$$

$$E_b = \frac{P\phi N}{60} \times \frac{Z}{A}$$



Types of DC Motor

- Series
- Shunt
- Compound

DC Series Motor

Field winding is connected in series with the motor

$$I_L = I_{SE} = I_a$$

Terminal voltage $V = E_b + I_a R_a + I_{SE} R_{SE}$

$$V = E_b + I_a R_a + I_{SE} R_{SE}$$

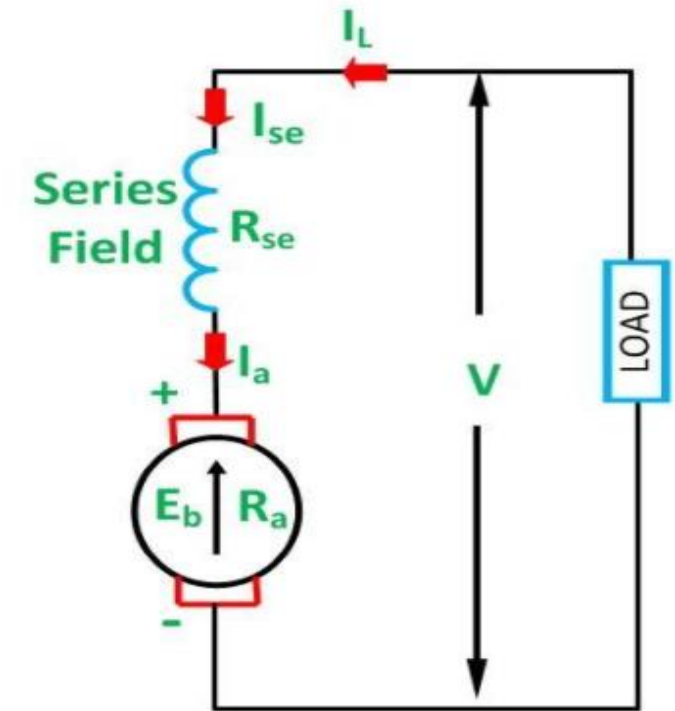
$$V = E_b + I_a (R_a + R_{SE})$$

$$E_b = V - I_a (R_a + R_{SE})$$

$$E_b = \frac{P\phi N}{60} \times \frac{Z}{A}$$

Z, A, P are constant

$$V - I_a (R_a + R_{SE}) = \frac{P\phi N}{60} \times \frac{Z}{A}$$



DC Shunt Motor

$$I_L = I_a + I_{Sh}$$

$$I_a = I_L - I_{Sh}$$

$$I_{Sh} = \frac{V}{R_{Sh}}$$

$$V = E_b + I_a R_a$$

$$V = E_b + (I_L - I_{Sh}) R_a$$

$$\text{Hence, } (I_L - I_{Sh}) R_a$$

$$E_b = V - (I_L - I_{Sh}) R_a$$

Compound Wound

Short Shunt

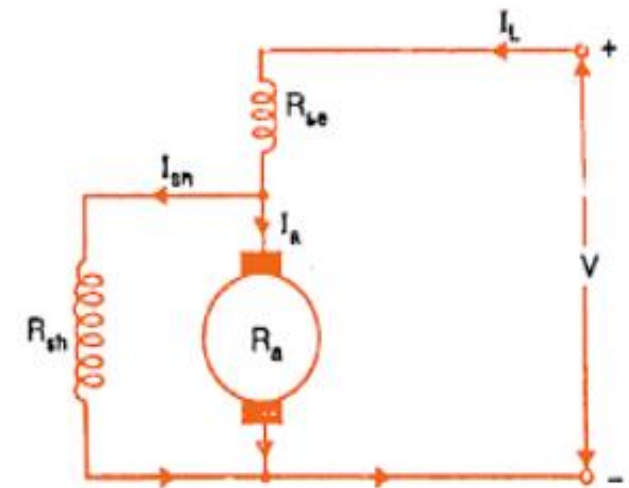
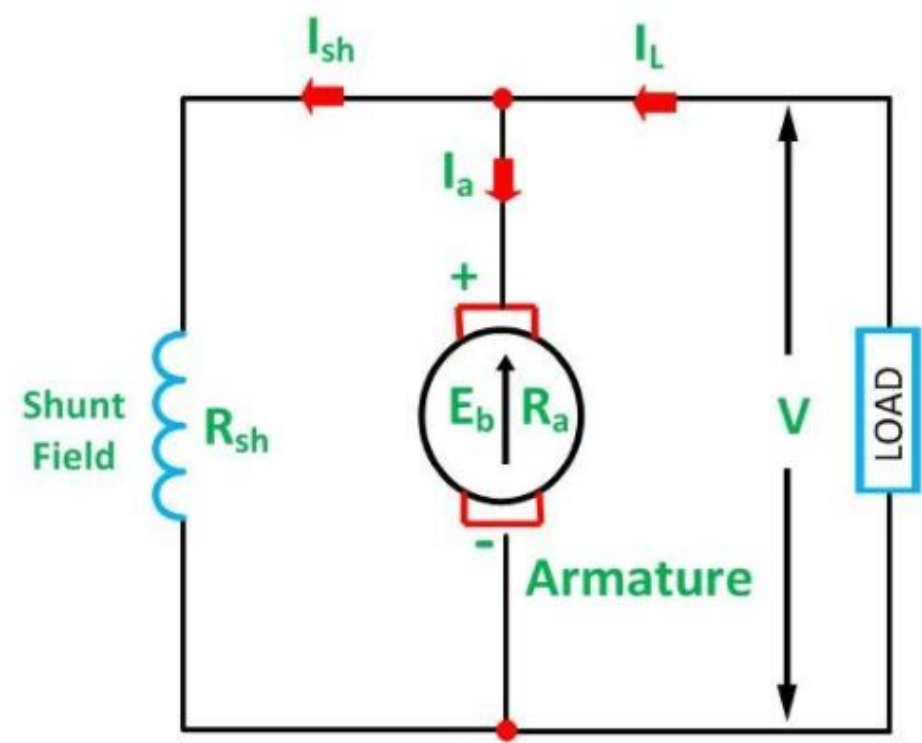
$$I_L = I_{SE} = I_a + I_{Sh}$$

$$I_a = I_{SE} - I_{Sh}$$

$$I_{Sh} = \frac{(V - I_{SE} R_{SE})}{R_{Sh}}$$

$$V = E_b + I_a R_a + I_{SE} R_{SE}$$

$$E_b = V - I_a R_a - I_{SE} R_{SE}$$



Long Shunt

$$I_L = I_{SE} + I_{Sh}$$

$$I_{SE} = I_a$$

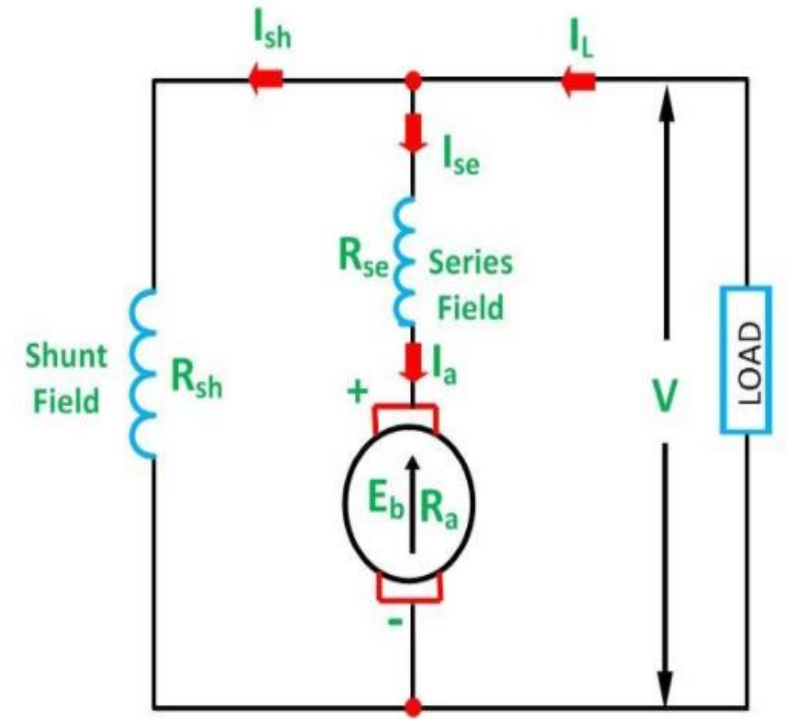
$$I_L = I_a + I_{Sh}$$

$$I_{Sh} = \frac{V}{R_{Sh}}$$

$$V = E_b + I_a R_a + I_{SE} R_{SE}$$

$$V = E_b + I_a (R_a + R_{SE})$$

$$E_b = V - I_a (R_a + R_{SE})$$



Comparison between shunt, series and compound wound DC Motor

Type of motor	Characteristics	Applications
shunt	Approximately constant speed. Speed can be controlled. Medium starting torque (Upto 1.5 Full load torque)	For driving constant speed line shafting lathes, centrifugal pumps, machine tools, blowers and fans, reciprocating pumps.
Series	Variable speed. Speed can be controlled. High starting torque.	For traction work i.e. electric locomotives rapid transit systems trolley cars etc. cranes and hoists, conveyors
Cumulative compound	Variable speed Speed can be controlled. High starting torque.	For intermittent high torque loads, for shears and punches, elevators, conveyors, heavy planers, rolling mills, ice machines, printing presses, air compressors

Why Series motor is called high starting torque motor?

Torque $T \propto \phi I_a$

$\phi \propto I_{SE}$

$I_L = I_{SE} = I_a$

In series motor $I_{SE} = I_a$

$\phi \propto I_a$

$T \propto I_a^2$

Torque is proportional to the square of the current

Hence there is a high starting torque

Why Shunt motor is called constant speed motor?

$$V = E_b + (I_L - I_{Sh}) R_a$$

$$E_b = \frac{P\phi N}{60} \times \frac{Z}{A}$$

P,Z,A are constant

$$E_b \propto \phi N$$

$$V \propto \phi N$$

Flux ϕ in shunt motor depends on current through the shunt windings

$$I_{Sh} = \frac{V}{R_{Sh}} \quad V = \text{Terminal Voltage}$$

V = constant, i.e, Terminal voltage

R_{Sh} = constant , shunt resistance

Hence I_{Sh} is constant

$$\phi \propto I_{Sh}$$

Shunt motor is called constant speed motor

INDUCTION MOTOR

When supply given to the motor is ac then it is called ac motor

2 types

1. Synchronous Motor
2. Asynchronous Motor or Induction Motor

Advantages of Induction motor

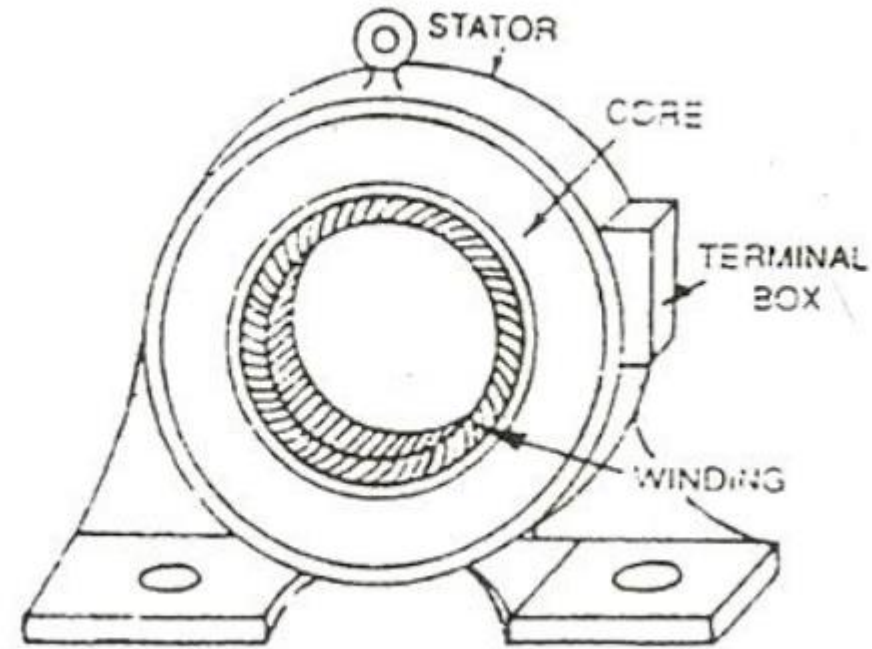
- Induction motor is widely used
- It has self starting capacity
- Heavy load capacity
- Simple design
- Can be used in any environmental conditions

Construction

Two major parts

Stator

Rotar



Stator

High grade steel alloy

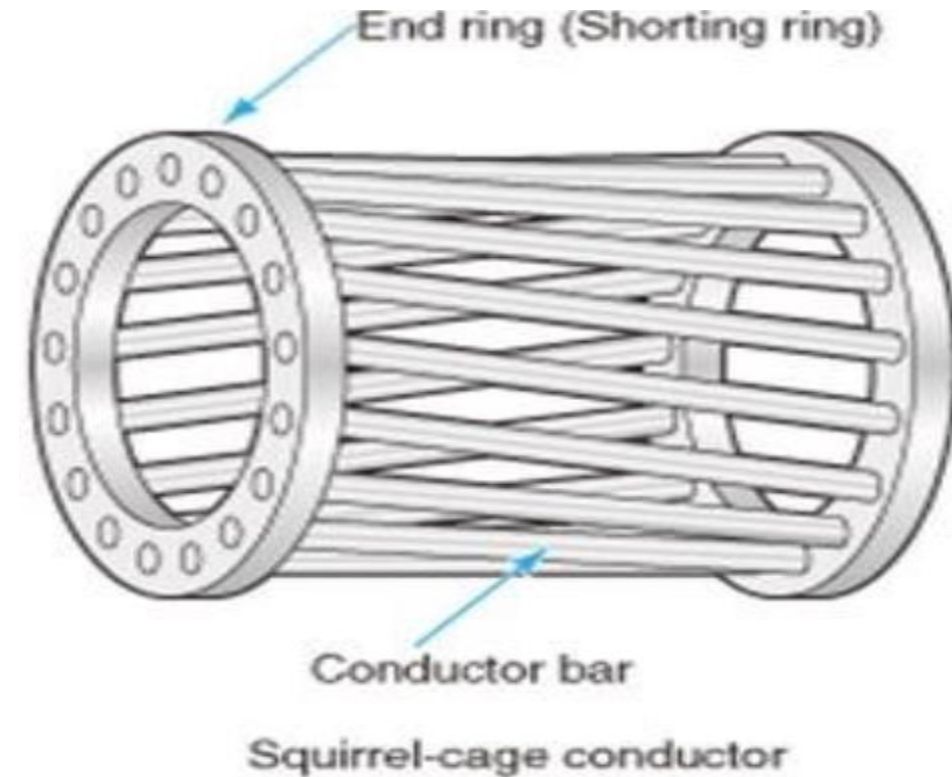
- Frame – Outerbody
- Core: Laminations of High Grade silicon Steel
Thickness 0.3 to 0.5mm
Inner side of the stamping slots are punched
Core carries the magnetic field

Rotar

- Squirrel cage
- Slip Ring/ Wound Rotar

Squirrel Cage

- Laminated cylindrical core type
- Conductors are in form of aluminum or copper bars, all bars are connected to slots and are connected to end rings
- Since it is shorted at the end rings external resistance or circuit cannot be added to it
- Slots are not exactly parallel they are squeed in order to reduce humming, rotar locking etc.
- Advantages: Cheaper, simple construction, Less maintenance, High efficiency



Wound Rotar / Slip ring Rotar

- Slip rings and brushes available
- External resistance can be added, Hence it will have high starting torque
- Low starting Current
- Speed control will be easy

Working

Asynchronous motor

Speed will ne less than N_s , Synchronous speed

Synchronous speed : speed of rotation of magnetic filed in a rotatory machine

$$N_s = \frac{120f}{P}$$

Works on the principle of electro magnetic induction

Induction motor is also called as rotating transformer.

SINGLE PHASE INDUCTION MOTOR

Most widely used in domestic as well as industrial purposes

Economic, Cheaper, reliable and easy maintenance

Used in : Washing machine compressor, blower, Home appliances

Construction

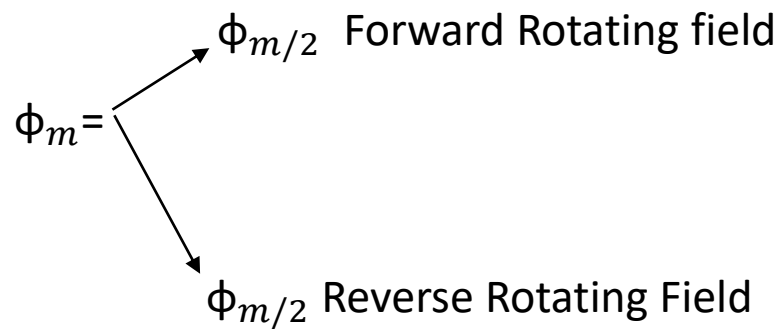
Almost similar to 3 ϕ Induction motor

Working

1 ϕ supply is provided hence the flux ϕ_m produced will be alternating ,not rotating

Double revolving theory

Alternating flux ϕ_m divides into two



- Flux produced in stator will not reach the rotar
- For starting,an additional flux with phase shift is provided thus changing the alternating nature to rotating

There are different methods to start an induction motor

Based on the method of starting the induction motor is classified into different types

1. Split phase induction motor
2. Capacitor start induction motor
3. Capacitor start capacitor Run induction motor
4. Shaded pole Induction Motor
5. Permanent split capacitor motor

Applications

ALTERNATOR

Generates an alternating voltage

Commutator is not required

Requires

Field

Conductor

Either field or conductor should be rotating

In order to generate a 3 ϕ voltage three conductors placed 120⁰ apart

Difference from DC generator

Conductor is stationary

Field is rotating

DC excitation is provided to the rotar to generate the field and this field rotates

When field rotates conductor cuts the flux then EMF is induced

The generated EMF is taken out from the stator

Construction

Stationary part

Stator Frame : Provide mechanical support

Stator Core : Several slots are present windings are provided in this slots

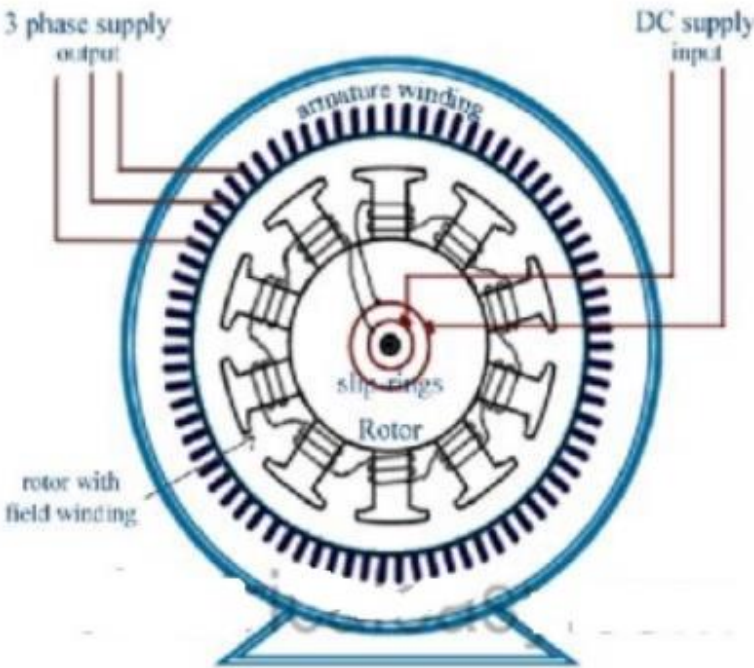
Laminated using Iron and silicon steel to reduce eddy current losses

O/P is taken to the supply

Rotar: 2 types

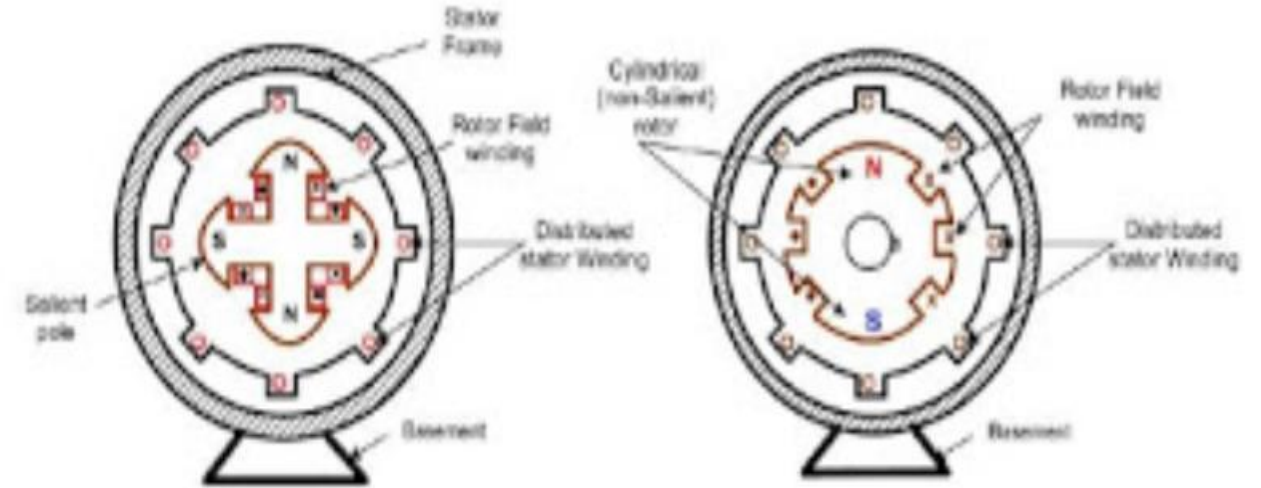
Salient pole

Cylindrical



Salient pole

- Projected ends
- Field windings are wound on this projected ends
- Low and medium speed applications
- Noise is high
- Cannot withstand high mechanical stress
- Non uniform air gap



Cylindrical

- Slots are available at periphery conductors are wound in this slots
- Consist of solid forged steel cylinder
- High speed applications
- Uniform airgap
- Less noise
- Highly Efficient

Working

- An alternator operates on the same fundamental principle of electromagnetic induction as a DC generator.
- The working of an alternator is based on the principle that when the flux linking a conductor changes, an emf is induced in the conductor.
- The rotor winding is energized from the DC exciter and alternate N and S poles are developed on the rotor.
- When the rotor is rotated in the anti-clockwise direction by a prime mover, the stator or armature conductors are cut by the magnetic flux of rotor poles.
- Consequently, e.m.f. is induced in the armature conductors due to electromagnetic induction
- The direction of induced e.m.f. can be found by **Fleming right-hand rule**

What is the advantage of using the rotating field in ac generator

TRANSFORMER

- Static device that transfers energy from one electrical circuit to another without change in frequency through magnetic medium
 - Winding that receives energy – Primary winding
 - Winding that delivers energy – secondary winding
- Based on voltage levels transformers are classified into
 - Step up Transformer
 - Step down Transformer

Construction

Depending upon the manner in which the primary and secondary windings are placed on the core and shape of the core there are two type of transformers

- Core type
- Shell type

Core Type

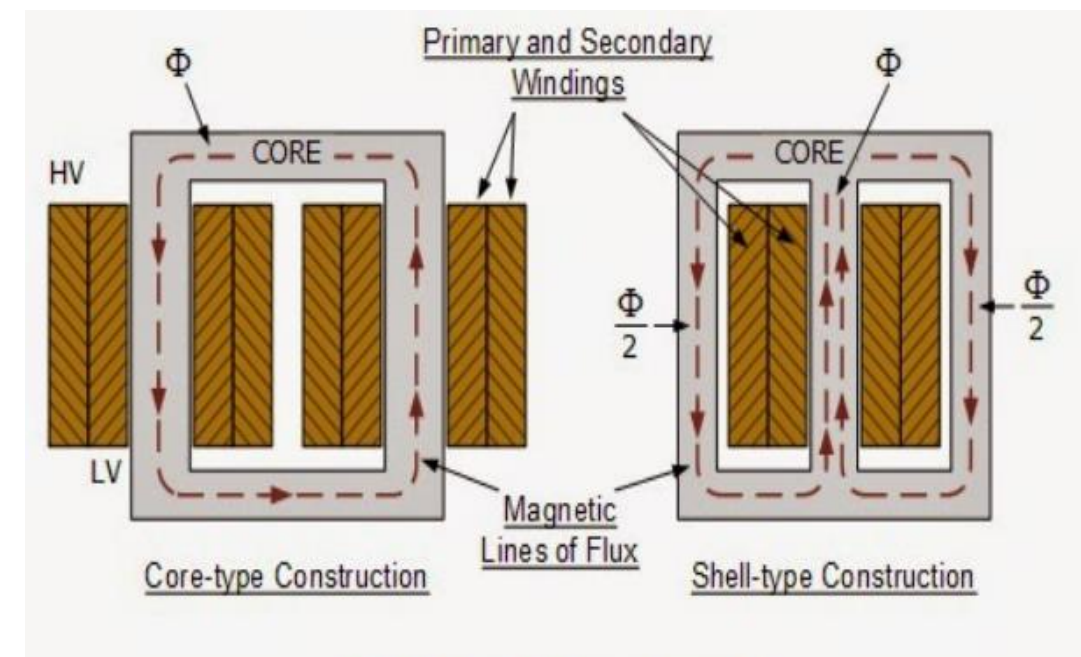
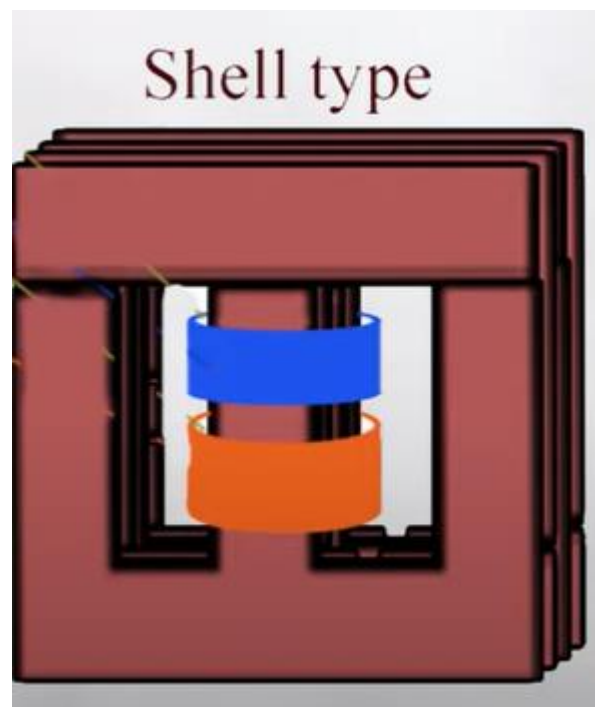
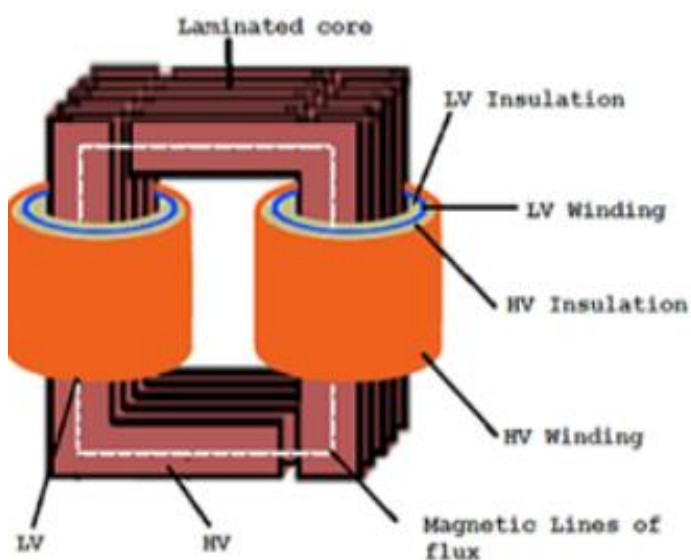
Windings are placed in the form of concentric cylindrical coils placed around the vertical limbs of the core.

The low-voltage (LV) as well as the high-voltage (HV) winding are made in two halves, and placed on the two limbs of core.

Shell Type

In the shell type transformer, the primary and secondary windings are wound over the central limb of a three-limb core

The HV and LV windings are split into a number of sections, and the sections are interleaved or sandwiched i.e. the sections of the HV and LV windings are placed alternately.



The core is built-up of thin steel laminations insulated from each other. This helps in reducing the eddy current losses in the core. The material commonly used for core is CRGO (Cold Rolled Grain Oriented) steel.

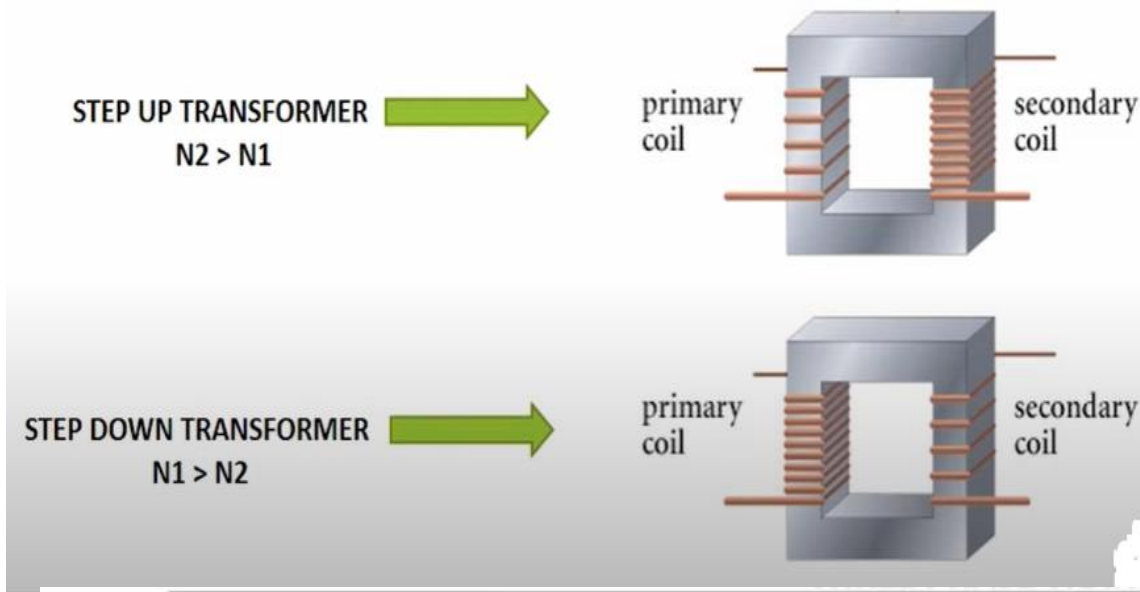
INSULATING OIL -In oil-immersed transformer, the iron core together with windings is immersed in insulating oil. The insulating oil provides better insulation, protects insulation from moisture and transfers the heat produced in core and windings to the atmosphere.

TANK AND CONSERVATOR- The transformer tank contains core wound with windings and the insulating oil. Conservator provides space when insulating oil expands due to heating. The transformer tank is provided with tubes on the outside, to permit circulation of oil, which aids in cooling. Some additional devices like breather and Buchholz relay are connected with main tank.

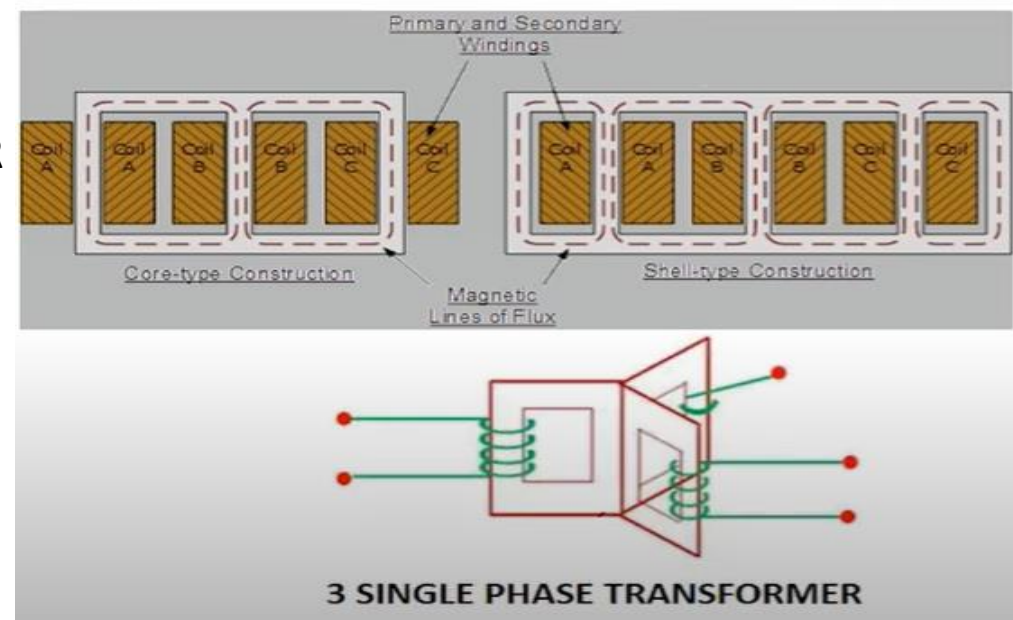


CLASSIFICATION OF TRANSFORMER

1 STEP UP AND STEP DOWN



2.BASED ON SUPPLY – THREE PHASE AND SINGLE PHASE TRANSFORMER



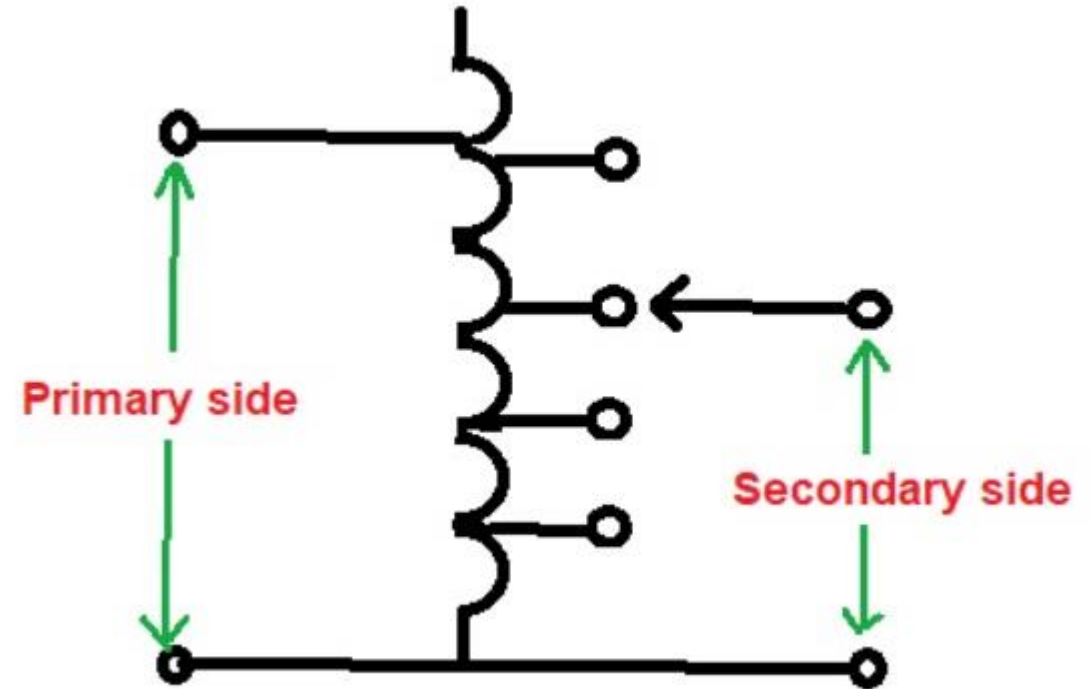
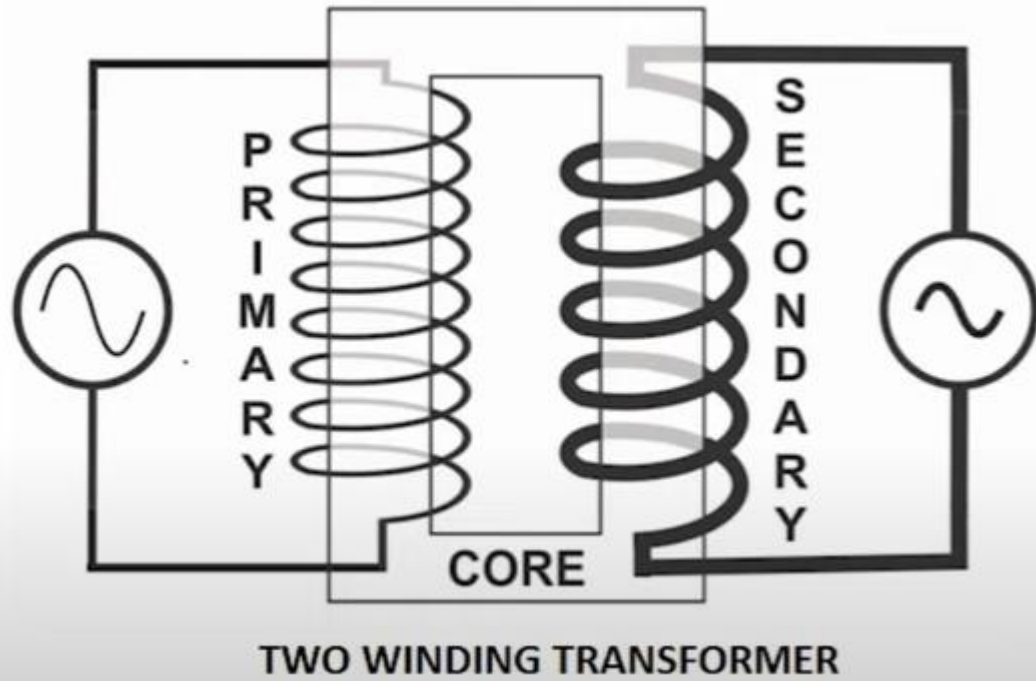
CLASSIFICATION OF TRANSFORMER

3 POWER, DISTRIBUTION AND INSTRUMENT TRANSFORMER



CLASSIFICATION OF TRANSFORMER

4 TWO WINDING AND AUTO TRANSFORMER



CLASSIFICATION OF TRANSFORMER

5 OUTDOOR AND INDOOR TRANSFORMER

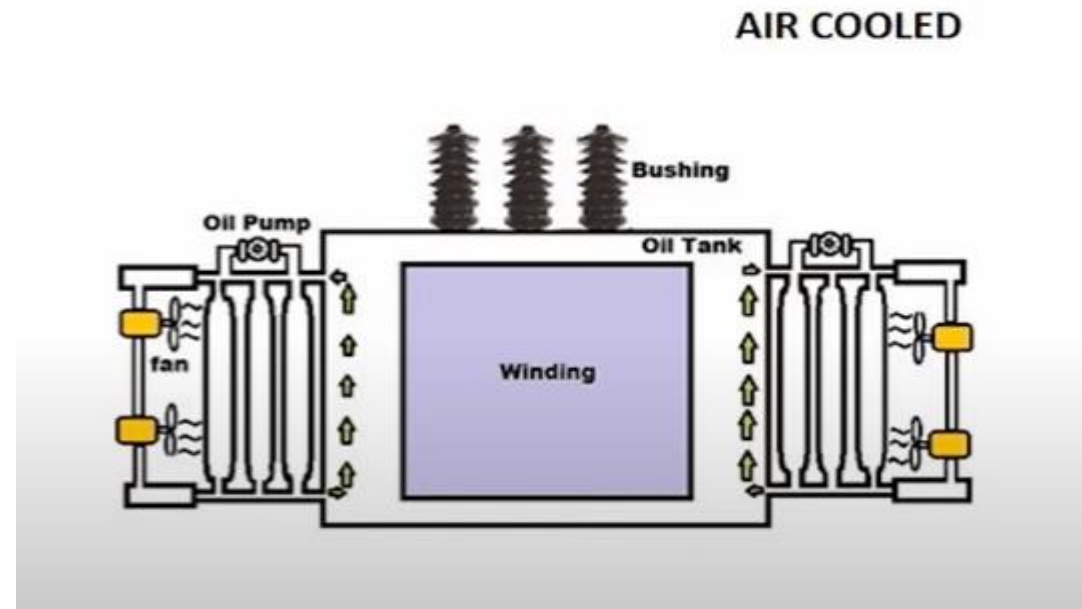
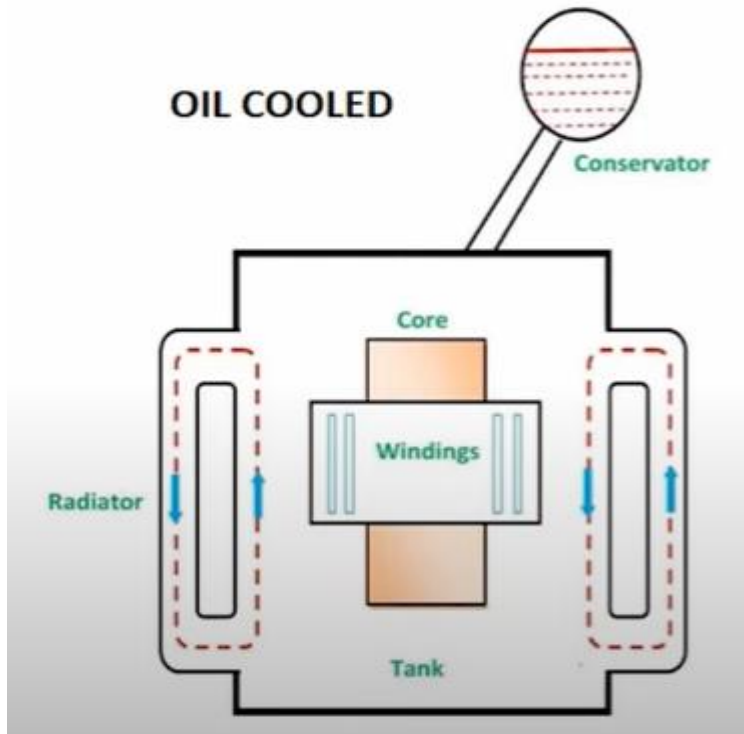


OUTDOOR TRANSFORMER



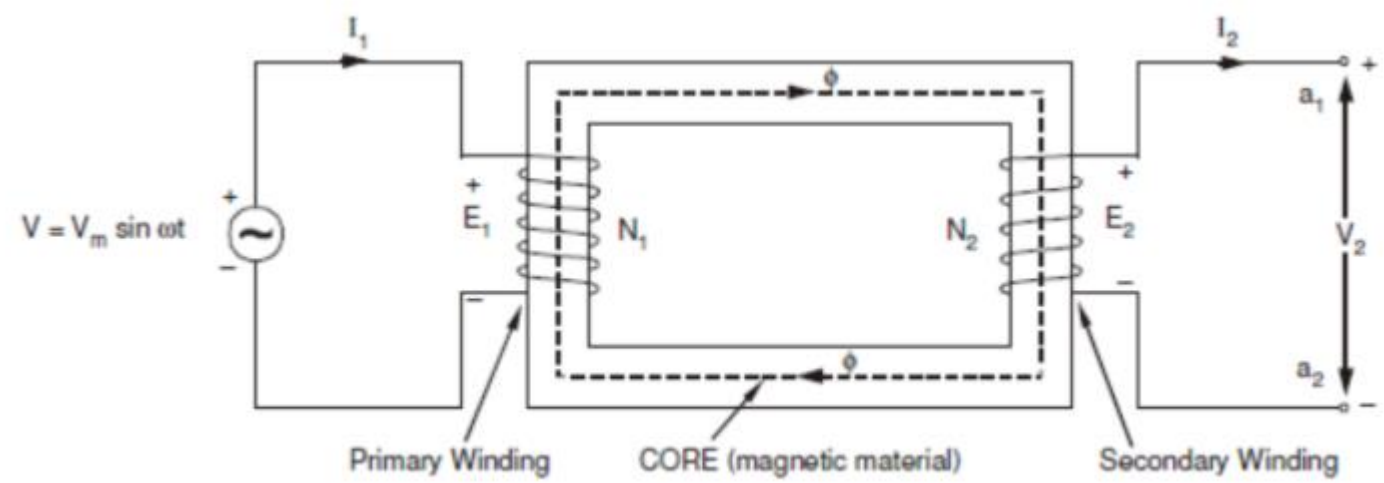
CLASSIFICATION OF TRANSFORMER

6 OIL COOLED AND DRY TYPE TRANSFORMER



7. CORE TYPE AND SHELL TYPE

Working Principle



What happens when transformer is supplied by DC?

VARIOUS LEVELS OF POWER TRANSMISSION

